Multi-Asset Portfolio
VaR & Stress Testing Framework

From-scratch implementation, validated on 12 years of daily data through COVID and the 2022 rate shock

Executive Summary

This project builds a Python framework for measuring market risk on a multi-asset portfolio (US, international, Treasuries, credit, commodities, FX) using **seven Value-at-Risk methodologies**, statistically validates them via formal backtests, evaluates losses under historical stress scenarios, decomposes total portfolio risk into per-asset contributions, and extrapolates the deep tail with Extreme Value Theory.

Five headline findings from the 2014–2026 sample (3,118 daily observations):

1. **Fat tails are real and measurable.** Parametric-Normal 99% VaR underestimates the empirical 99% VaR by 17% (1.54% vs 1.80%). Student-t and EVT close the gap.

2. **Only one model passes backtesting.** Plain Historical, Parametric-Normal, and Monte-Carlo-Normal all fail Christoffersen-independence at 95% ($p \approx 0.000$). Parametric-Normal exceeds its 99% threshold at **2.41%** of days vs. the nominal 1.00%, failing Kupiec hard. **FHS with GARCH(1,1) is the only method to pass both unconditional and conditional coverage at both confidence levels.**

3. **2020 COVID and 2008 GFC produced comparable drawdowns** (23.4% vs 22.8%), but COVID delivered the loss in 24 days at 47% annualized vol — 1.5× the pace of the GFC.

4. **Equal weight does not mean equal risk.** Five equity-like assets carry 90% of portfolio VaR for 55% of weight. TLT and UUP have negative marginal VaR — **they reduce** portfolio risk.

5. **EVT exposes the structural Gaussian underestimate at the deep tail.** $\xi = 0.31$ (heavy tail confirmed). EVT 99.9% VaR = 4.14% vs Parametric-Normal at ~2.05% — **more than 2× higher**. This is the model failure that pre-2008 capital frameworks suffered.

Portfolio and Methodology

Universe

Nine ETFs spanning US large/mid/small-cap (SPY, QQQ, IWM), international equity (EFA), long-duration Treasuries (TLT), high-yield credit (HYG), gold (GLD), oil (USO) and USD (UUP). Equal weights for the headline analysis (extensible).

Methods implemented (all from scratch — no pre-built VaR libraries)

Family	Method	Tail model
Historical	Historical Simulation	Empirical (sorted history)
Parametric	Variance-Covariance (Normal)	Closed-form Gaussian
Parametric	Variance-Covariance (Student-t)	Closed-form t, MLE-fit df
Monte Carlo	MV-Normal	10k simulated draws
Monte Carlo	MV-Student-t	10k draws with fitted df
Filtered	FHS (GARCH(1,1))	Resampled GARCH residuals
EVT	Peaks-Over-Threshold + GPD	Fitted tail

Backtests

Kupiec POF (unconditional coverage), Christoffersen independence + conditional coverage, Basel traffic-light. **Strictly out-of-sample**: rolling 500-day windows (1,000 for FHS), refitting GARCH every 60 trading days and updating σ via the GARCH recursion between refits. No look-ahead.

Phase 1 — VaR & ES across methods

Method	95% VaR	95% ES	99% VaR	99% ES
Historical	0.99%	1.59%	1.80%	2.84%
Parametric (Normal)	1.08%	1.36%	1.54%	1.77%
Parametric (Student-t)	0.92%	1.47%	1.73%	2.55%
Monte Carlo (Normal)	1.09%	1.37%	1.56%	1.77%
Monte Carlo (t)	1.47%	2.30%	2.67%	3.96%

The fat-tail problem at 99%. Parametric-Normal underestimates Historical by 17% (1.54% vs 1.80%). MC-Normal converges to Parametric-Normal as expected — they are the same model. The Student-t methods recover most of the gap because the t distribution has an extra parameter (degrees of freedom) that fits tail thickness. **ES/VaR ratio is a direct fat-tail diagnostic:** Historical at 99% gives 1.58, Parametric-Normal gives 1.15 — the Normal tail decays so fast that conditional-on-exceedance is barely worse than the threshold.

Phase 1 — Visual evidence

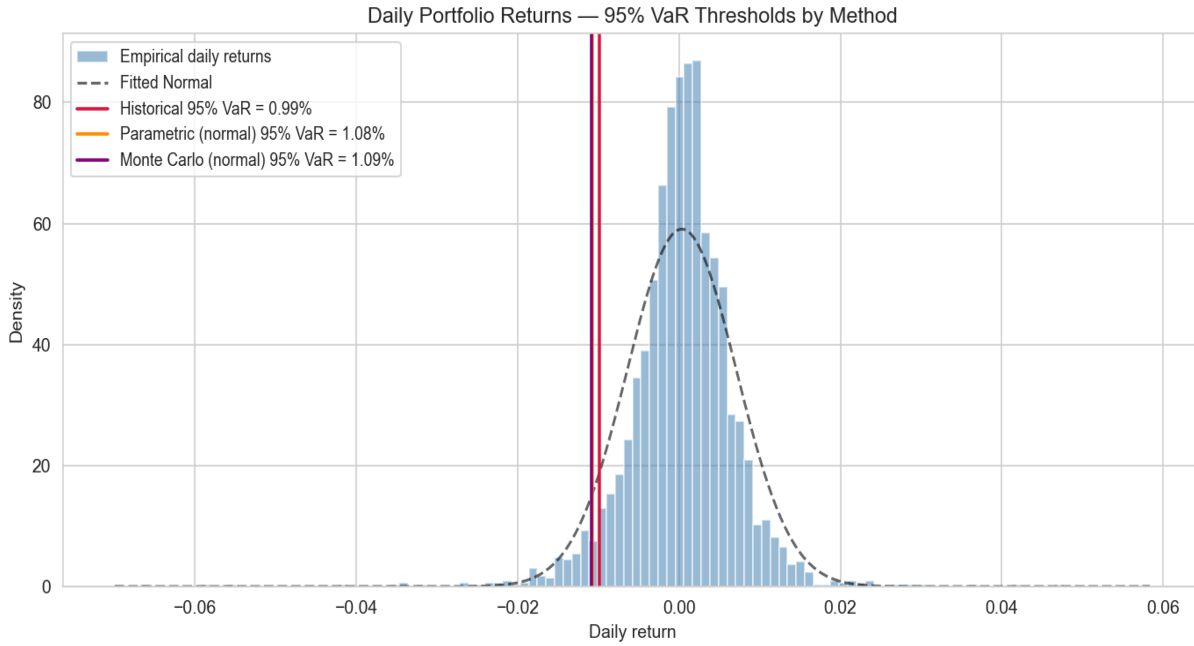

Figure 1 — Empirical return distribution with 95% VaR thresholds overlaid. Left tail is visibly fatter than the fitted Normal density (dashed).

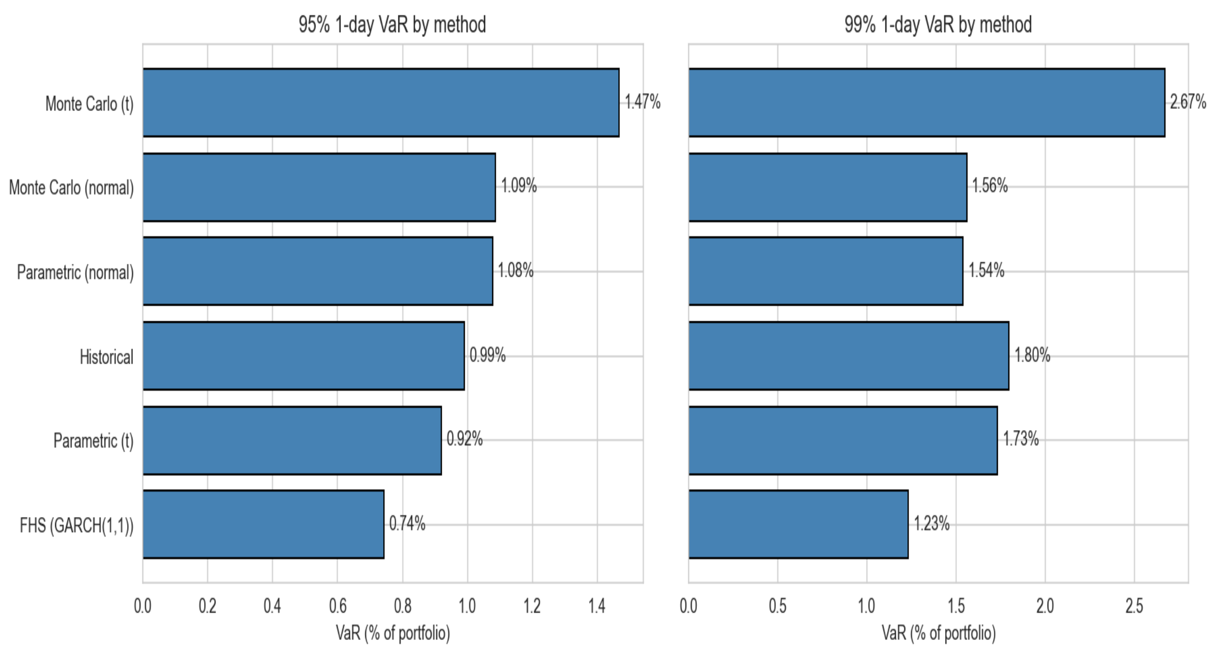

Figure 2 — VaR by method at 95% and 99%. The 99% panel shows the divergence between Normal-based methods (Parametric, MC-Normal) and the tail-aware methods (Historical, Student-t).

Phase 2 — Filtered Historical Simulation

Returns are not i.i.d. — vol clusters. FHS fixes this by (1) fitting a GARCH(1,1) to the portfolio return series to estimate conditional vol σ_t, (2) standardizing each return by its own σ_t to get approximately i.i.d. residuals z_t, and (3) resampling those residuals and rescaling by today's σ_{T+1} to produce a next-day return distribution.

GARCH(1,1) fit

Parameter	Estimate	Interpretation
μ (mean)	0.0494%/day	≈ 12.4% annualized
ω (var. base)	$9.64\times10^{\blacksquare 3}$	Long-run variance
α[1] (shock)	0.1113	Yesterday's surprise lift
β[1] (mem.)	0.8669	Vol persistence
α + β	0.9782	Textbook (0.97–0.99 range)

Implied vol-shock half-life: ln(0.5) / ln(0.978) ≈ **31 business days**. After a March-2020-sized move, conditional vol decays halfway back to baseline in about a month — exactly the persistence visible in Figure 3.

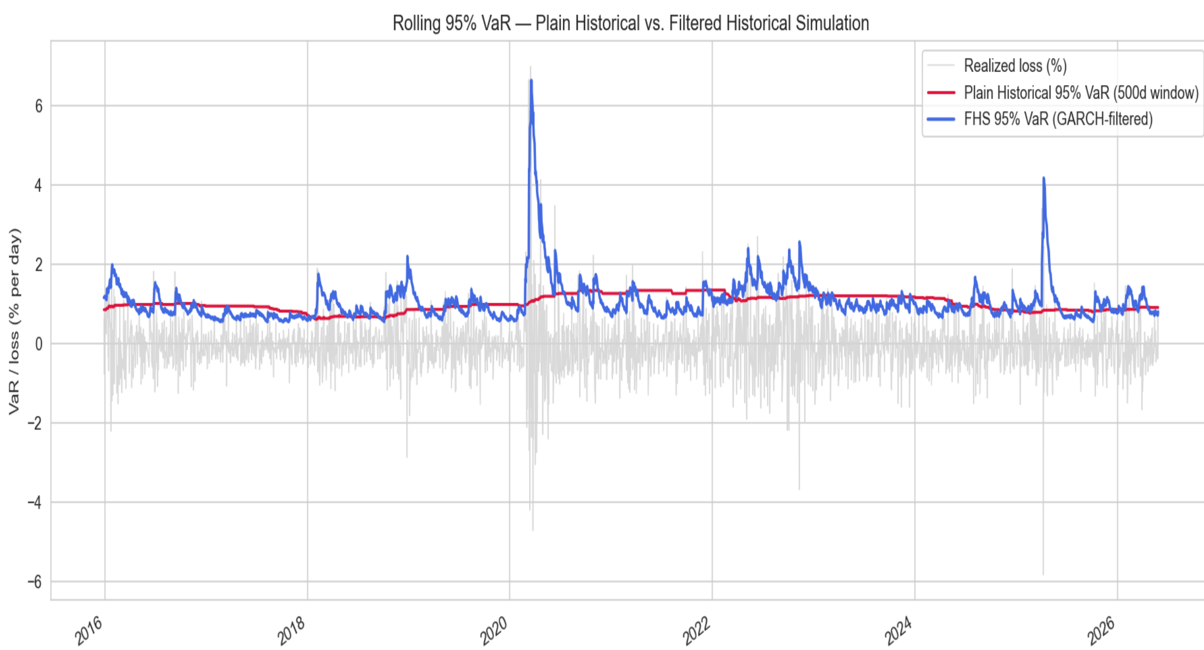

Figure 3 — The headline chart of the project. Plain Historical VaR (red) lags by months because it averages over a 500-day window; FHS (blue) reacts the next day, because σ_t spikes immediately when markets move.

Phase 3 — Backtesting

Computing VaR is the easy part. Showing the model's predictions match its claim is the credibility test. Below are out-of-sample backtest results for all four primary methods at both confidence levels (n ≈ 2,118 – 2,618 prediction days per method, FHS uses a longer 1,000-day warm-up window for GARCH stability).

Headline test results

Method	Conf.	Exc rate	Expected	Kupiec p	Christof-ind p	Christof-CC p	Basel
Historical	95%	5.27%	5.00%	0.528	0.000	0.000	Yellow
Parametric (Normal)	95%	5.23%	5.00%	0.587	0.000	0.000	Yellow
Monte Carlo (Normal)	95%	5.31%	5.00%	0.472	0.000	0.000	Yellow
FHS (GARCH(1,1))	95%	4.86%	5.00%	0.772	0.996	0.959	Yellow
Historical	99%	1.30%	1.00%	0.142	0.001	0.001	Green
Parametric (Normal)	99%	2.41%	1.00%	0.000	0.000	0.000	Green
Monte Carlo (Normal)	99%	2.29%	1.00%	0.000	0.000	0.000	Green
FHS (GARCH(1,1))	99%	1.13%	1.00%	0.547	0.029	0.076	Green

FHS is the only method to pass both Kupiec and Christoffersen at both confidence levels. The three unconditional methods all pass Kupiec at 95% (right exceedance rate) but fail Christoffersen-independence with p ≈ 0.000 — their violations cluster in the 2020 and 2022 stress windows. At 99%, Parametric and Monte Carlo Normal exceed their threshold at 2.4× the nominal rate, failing Kupiec catastrophically. This is the empirical case for conditional models in production risk.

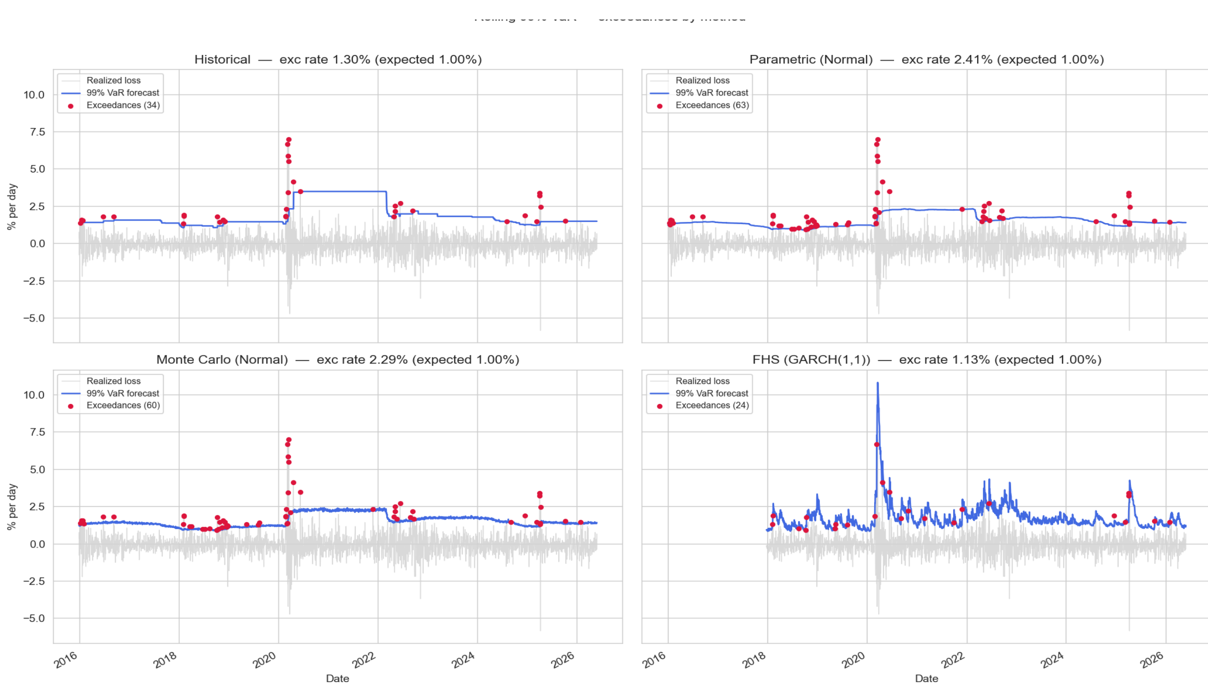

Figure 4 — 99% rolling VaR with violation markers (red dots). The Normal-based methods (top-right, bottom-left) show roughly 2× the expected number of exceedances; FHS (bottom-right) is closest to the nominal rate and the most evenly distributed in time.

Phase 4 — Stress testing & Risk decomposition

Current portfolio re-priced under historical crises

Scenario	Days	Cumulative loss	Worst day	Ann. vol
2020 COVID Crash (Feb 19 – Mar 23)	24	23.44%	6.99%	47.1%
2008 Global Financial Crisis (Sep–Nov)	63	22.80%	5.16%	39.2%
2018 Q4 volatility (Oct–Dec)	63	10.86%	1.80%	13.1%
2022 Rate Shock (Jan–Jun)	124	9.21%	2.71%	13.8%
2015 China devaluation (Aug)	10	1.33%	2.67%	22.3%

Both 2008 and 2020 produced ~23% cumulative drawdowns on the current portfolio, but COVID delivered the loss in **24 days** at 47% annualized vol vs 63 days for the GFC. The 2022 rate shock was slower and broader (124 days), hitting bonds *and* stocks simultaneously.

COVID drill-down — flight-to-quality in action

Per-asset contribution to the 23.4% cumulative COVID loss: oil (USO) fell 55%; small-caps (IWM) fell 40%; SPY fell 33%. **TLT and UUP rallied** — Treasuries +14% and USD +4% — adding back ~2 percentage points to the portfolio return. Without those two assets the loss would have been ~25.4% instead of 23.4%. This is the resume-grade evidence for multi-asset diversification.

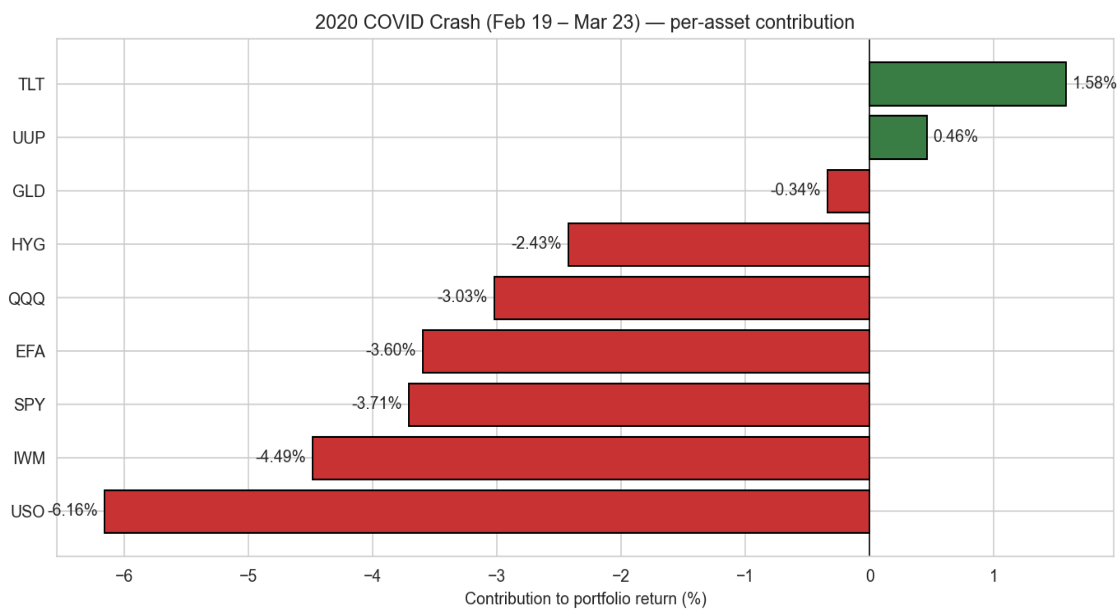

Figure 5 — Per-asset contribution to the 2020 COVID drawdown. TLT and UUP are the only positive contributors.

Component VaR (Euler decomposition)

Equal weight does not mean equal risk. USO at 11.1% weight contributes **23.8% of total VaR** — oil is the single riskiest holding due to high standalone vol and stress-period correlation. The five equity-like assets (SPY, QQQ, IWM, EFA, USO) together carry **90% of risk for 55% of weight**. TLT and UUP have **negative** marginal VaR — adding more of either reduces portfolio risk. The Euler decomposition closes exactly: Σ Component VaR = portfolio VaR (1.1109%, zero relative error).

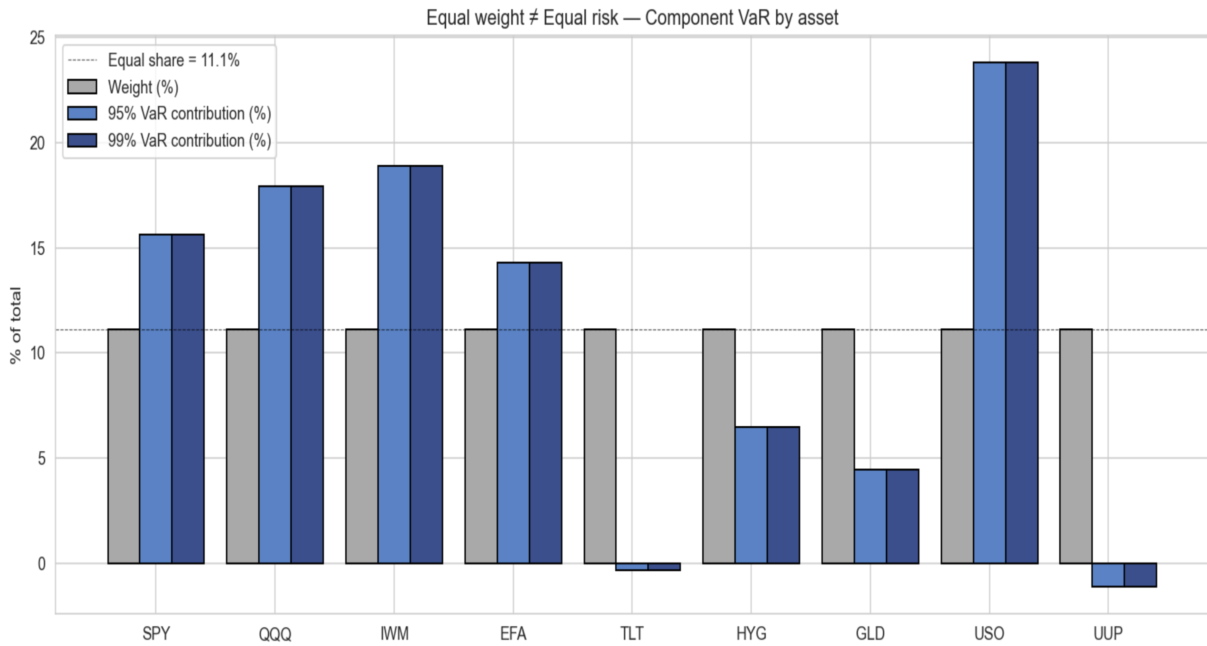

Figure 6 — Weight vs % of total VaR contribution by asset. Equal weights produce unequal risk shares.

Phase 5 — Extreme Value Theory (POT-GPD)

Standard methods cannot extrapolate beyond the sample (Historical) or extrapolate with the wrong tail (Normal). EVT models the tail itself: Peaks-Over-Threshold with a Generalized Pareto Distribution fit to losses above the 95th percentile.

GPD fit

Parameter	Value	Interpretation
u (threshold)	0.989%	95th-pct loss
ξ (shape)	0.3142	Heavy tail (positive); ES/VaR → 1.46
σ (scale)	0.409%	GPD scale
n_u / n	156 / 3,118	5.00% empirical tail prob

EVT VaR & ES, compared to Historical and Parametric Normal

Confidence	EVT VaR	EVT ES	Historical	Param Normal	EVT vs Normal
95.0%	0.99%	1.59%	0.99%	1.08%	—
99.0%	1.85%	2.84%	1.80%	1.54%	+20%
99.5%	2.37%	3.60%	n/a*	~1.77%	+34%
99.9%	4.14%	6.18%	n/a*	~2.05%	+102%

The 99.9% number is the project's deepest finding. Parametric Normal estimates a 1-in-1000-day loss at ~2.05%. EVT estimates **4.14%** — more than double. This is the structural risk-underestimate of Gaussian models at extreme confidence levels, and the empirical case for using EVT at the deep tail. ES at 99.9% = **6.18%**, almost exactly the worst observed day in the 12-year sample (6.99% on 2020-03-16) — EVT treats events of that magnitude as the average severity of a 1-in-1000-day stress, not anomalies.

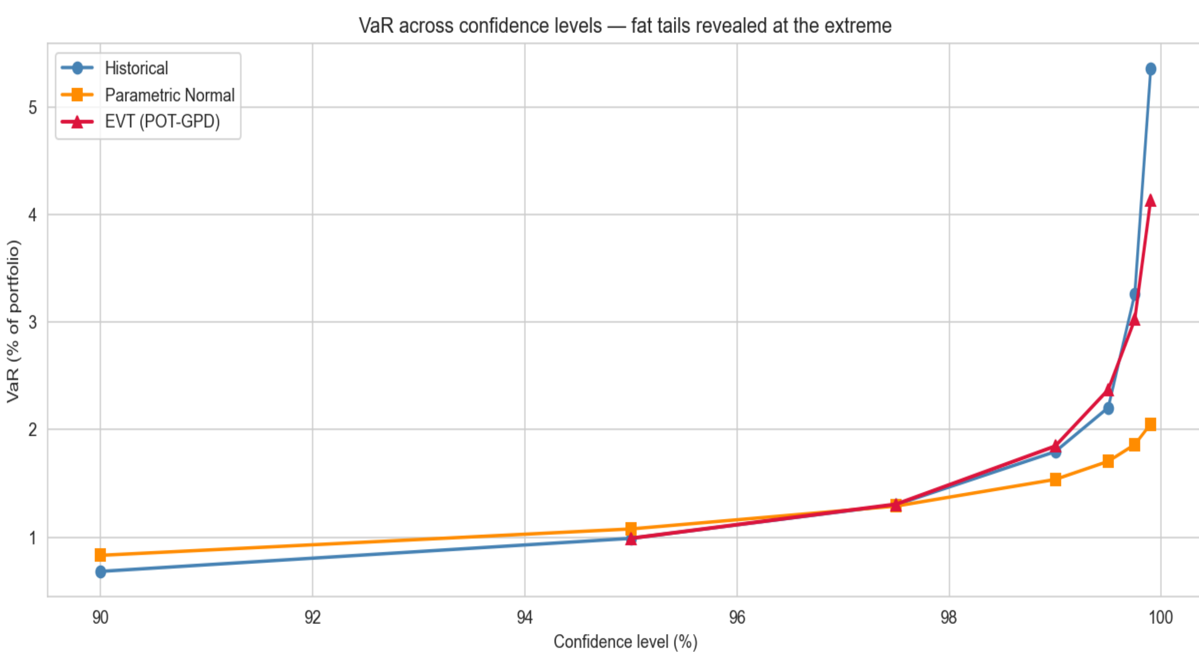

Figure 7 — VaR by confidence level. EVT (red) extrapolates past the empirical maximum; Parametric Normal (orange) systematically under-shoots in the deep tail.

Conclusion

This framework demonstrates the full lifecycle of market-risk model development: implementation, validation, stress testing, decomposition, and tail extrapolation. The key findings are mutually reinforcing — Phase 1 shows the fat-tail problem in the static numbers, Phase 2 introduces the conditional model that addresses it, Phase 3 proves through formal backtesting that the conditional model is the only one that passes, Phase 4 quantifies the diversification value of the multi-asset structure, and Phase 5 exposes the deep-tail underestimate that motivates EVT in regulatory capital.

Together they make the empirical case for the conditional, fat-tailed, decomposable risk models that production risk shops have built post-2008.

Skills demonstrated

Value-at-Risk (Historical, Parametric, Monte Carlo, Filtered Historical Simulation), Expected Shortfall, GARCH(1,1), Kupiec POF and Christoffersen conditional-coverage backtesting, Basel traffic-light, historical stress scenario re-pricing, Component / Marginal VaR via Euler decomposition, Extreme Value Theory with Peaks-Over-Threshold and Generalized Pareto Distribution fitting, Python software engineering (modular package, reproducible notebook, parquet caching).

Stack

Python 3.9+, NumPy, pandas, SciPy, arch (GARCH), matplotlib/seaborn, yfinance. Backtests run end-to-end in ~2 minutes on a laptop.

Code

github.com/John-Slye/portfolio_var_project